

ABDELHAMED FATHY

Full-Stack Software Engineer (Laravel & Vue.js)

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Full-Stack Software Engineer with strong hands-on expertise in PHP (Laravel), Vue.js, and AI-driven web applications. Proven track record of architecting production-grade platforms, real-time communication systems, and open-source packages focusing on automated database optimization and AI integration (Gemini API, RAG, FastAPI). Passionate about scalable architecture, high-performance web applications, and solving complex backend challenges.

TECHNICAL SKILLS

Backend & Architecture: PHP (Laravel), Python (FastAPI), RESTful API Design, Microservices, Event-Driven Architecture, Queue-Based Job Processing

Frontend: Vue.js 3 (Composition API), Inertia.js, Pinia, Blade, Tailwind CSS, JavaScript (ES6+), HTML5/CSS3

Real-Time & Performance: WebSockets (Laravel Reverb, Pusher), WebRTC Signaling, Redis (Queues, Caching & Presence Channels), MySQL Query Optimization

AI Integration: Google Gemini API, OpenAI APIs, LangChain, RAG Systems, Vector Databases (ChromaDB), Custom NLP Model Training (scikit-learn)

DevOps & Tools: Docker & Docker Compose, Git, GitHub Actions, Jenkins CI/CD, Linux (Ubuntu/WSL2), Postman

PROJECTS

OptiQuery AI — Automated Database Optimization Platform [\[Demo Video\]](#) [\[GitHub\]](#) Jul 2026 – Sep 2026

Laravel · Vue.js 3 · Inertia.js · FastAPI · ChromaDB · Gemini AI · Docker

- Engineered a self-built Laravel client package that intercepts slow SQL queries and automates EXPLAIN-plan analysis via Gemini AI to diagnose root causes and suggest actionable indexes.
- Architected a FastAPI microservice integrated with ChromaDB powering a RAG-based assistant for conversational root-cause analysis and performance tuning.
- Delivered a Vue.js 3 + Inertia.js dashboard for visualizing query health trends and reviewing AI-suggested schema fixes in real time.

Bazzario — Multi-Vendor E-Commerce Platform [\[GitHub\]](#) May 2026 – Jun 2026

Laravel · Blade · FastAPI · Redis · Docker · Laravel Reverb

- Architected a multi-role marketplace (Admin, Vendor, Customer) with isolated dashboards, strict role-based permission rules, and robust order/inventory management.
- Integrated an LLM pipeline for automated product descriptions and built a customer-facing AI shopping assistant, optimizing real-time communications using Laravel Reverb and Redis.
- Designed the multi-tenant data layer to keep vendor stores fully isolated while sharing a single scalable codebase and deployment pipeline.

Nexus — Real-Time Chat & AI Communication Platform [\[Demo Video\]](#) [\[GitHub\]](#) Apr 2026 – Jun 2026

Laravel · WebSockets · WebRTC · Redis Queues · Gemini API

- Built an event-driven real-time chat platform supporting 1-to-1/group conversations, asynchronous voice message processing via queues, and instant push notifications.
- Implemented WebRTC signaling for low-latency peer-to-peer calls and embedded Gemini AI for context-aware, instant automated replies.
- Designed a Redis-backed queue architecture to offload media and AI processing, keeping the chat interface responsive under heavy load.

Several other production projects and experiments are available on my [GitHub profile](#) — [github.com/3bde17med](#)

OPEN-SOURCE PACKAGES

Sentinel AI — AI-Powered Laravel Exception Monitor [\[GitHub\]](#) 2026

Laravel · Google Gemini AI · Blade · Tailwind CSS

- Built an open-source Laravel package that automatically captures application exceptions and uses Gemini AI to generate root-cause explanations and fixes in Arabic and English.
- Shipped a zero-config, plug-and-play Blade + Tailwind dashboard for browsing, filtering, and clearing error logs with no extra frontend setup.

OptiQuery Laravel — Slow Query Reporting Client [\[GitHub\]](#) 2026

Laravel · MySQL · Async HTTP

- Published a Laravel client package that asynchronously reports slow SQL queries (Http::async(), configurable threshold) with zero performance overhead on the host app.
- Auto-captures EXPLAIN plans and table DDL per query for AI analysis, with built-in infinite-loop protection and zero-config Laravel package discovery.

MILITARY SERVICE

Completed mandatory military service — Egyptian Armed Forces, Air Defense Forces — with Honorable Discharge (Jan 2025 – Mar 2026).

EDUCATION

Bachelor's Degree in Electronic Engineering — Menoufia University

LANGUAGES

Arabic — Native

English — Professional Working Proficiency (Strong Reading & Writing)